## 4.3 Sine Function and Inverse Sine Function

Let us recall that sine function is a function with $\mathbb{R}$ as its domain and $[-1, 1]$ as its range. We write $y = \sin x$ and $y = \sin^{-1} x$ or $y = \arcsin(x)$ to represent the sine function and the inverse sine function, respectively. Here, the symbol $-1$ is not an exponent. It denotes the inverse and does not mean the reciprocal.

We know that $\sin(x + 2\pi) = \sin x$ is true for all real numbers $x$. Also, $\sin(x + p)$ need not be equal to $\sin x$ for $0 < p < 2\pi$ and for all $x$. Hence, the period of the sine function is $2\pi$.

### 4.3.1 The graph of sine function

The graph of the sine function is the graph of $y = \sin x$, where $x$ is a real number. Since sine function is periodic with period $2\pi$, the graph of the sine function is repeating the same pattern in each of the intervals, $\cdots$, $[-2\pi, 0]$, $[0, 2\pi]$, $[2\pi, 4\pi]$, $[4\pi, 6\pi]$, $\ldots$ . Therefore, it suffices to determine the portion of the graph for $x \in [0, 2\pi]$. Let us construct the following table to identify some known coordinate pairs for the points $(x, y)$ on the graph of $y = \sin x$, $x \in [0, 2\pi]$.

| $x$ ( in **radian** ) | 0 | $\frac{\pi}{6}$ | $\frac{\pi}{4}$ | $\frac{\pi}{3}$ | $\frac{\pi}{2}$ | $\pi$ | $\frac{3\pi}{2}$ | $2\pi$ |
|---|---|---|---|---|---|---|---|---|
| $y = \sin x$ | 0 | $\frac{1}{2}$ | $\frac{1}{\sqrt{2}}$ | $\frac{\sqrt{3}}{2}$ | 1 | 0 | $-1$ | 0 |

It is clear that the graph of $y = \sin x$, $0 \le x \le 2\pi$, begins at the origin. As $x$ increases from 0 to $\frac{\pi}{2}$, the value of $y = \sin x$ increases from 0 to 1. As $x$ increases from $\frac{\pi}{2}$ to $\pi$ and then to $\frac{3\pi}{2}$, the value of $y$ decreases from 1 to 0 and then to $-1$. As $x$ increases from $\frac{3\pi}{2}$ to $2\pi$, the value of $y$ increases from $-1$ to 0. Plot the points listed in the table and connect them with a smooth curve. The portion of the graph is shown in Fig. 4.4.

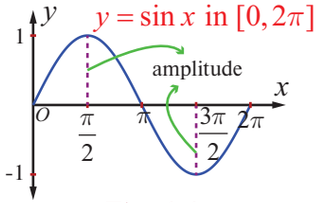


Fig. 4.4

**The entire graph of** $y = \sin x$, $x \in \mathbb{R}$ consists of repetitions of the above portion on either side of the interval $[0, 2\pi]$ as $y = \sin x$ is periodic with period $2\pi$. The graph of sine function is shown in Fig. 4.5. The portion of the curve corresponding to 0 to $2\pi$ is called a **cycle**. Its amplitude is 1.

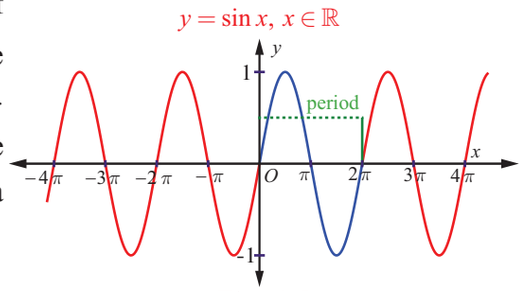


Fig. 4.5

**Note**

Observe that $\sin x \ge 0$ for $0 \le x \le \pi$, which corresponds to the values of the sine function in quadrants I and II and $\sin x < 0$ for $\pi < x < 2\pi$, which corresponds to the values of the sine function in quadrants III and IV.



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